

LEVEL OF SERVICE

LOS on basic freeway and multilane highway segments is defined by density. Although speed is a major concern of drivers related to service quality, describing LOS on the basis of speed would be difficult, since it remains constant up to high flow rates [i.e., 1,000 to 1,800 pc/h/ln for basic freeway segments (depending on the FFS) and 1,400 pc/h/ln for multilane highway segments]. Density describes a motorist's proximity to other vehicles and is related to a motorist's freedom to maneuver within the traffic stream. Unlike speed, density is sensitive to flow rates throughout the range of flows. Exhibit 12-14 illustrates the six levels of service defined for basic freeway segments.



LOS A



LOS B



LOS C



LOS D



LOS E



LOS F

Exhibit 12-14
LOS Examples for Basic
Freeway Segments

LOS Described

LOS A describes free-flow operations. FFS prevails on the freeway or multilane highway, and vehicles are almost completely unimpeded in their ability to maneuver within the traffic stream. The effects of incidents or point breakdowns are easily absorbed.

LOS B represents reasonably free-flow operations, and FFS on the freeway or multilane highway is maintained. The ability to maneuver within the traffic stream is only slightly restricted, and the general level of physical and

psychological comfort provided to drivers is still high. The effects of minor incidents are still easily absorbed.

LOS C provides for flow with speeds near the FFS of the freeway or multilane highway. Freedom to maneuver within the traffic stream is noticeably restricted, and lane changes require more care and vigilance on the part of the driver. Minor incidents may still be absorbed, but the local deterioration in service quality will be significant. Queues may be expected to form behind any significant blockages.

LOS D is the level at which speeds begin to decline with increasing flows, with density increasing more quickly. Freedom to maneuver within the traffic stream is seriously limited, and drivers experience reduced physical and psychological comfort levels. Even minor incidents can be expected to create queuing, because the traffic stream has little space to absorb disruptions.

LOS E describes operation at or near capacity. Operations on the freeway or multilane highway at this level are highly volatile because there are virtually no usable gaps within the traffic stream, leaving little room to maneuver within the traffic stream. Any disruption to the traffic stream, such as vehicles entering from a ramp or an access point or a vehicle changing lanes, can establish a disruption wave that propagates throughout the upstream traffic stream. Toward the upper boundary of LOS E, the traffic stream has no ability to dissipate even the most minor disruption, and any incident can be expected to produce a serious breakdown and substantial queuing. The physical and psychological comfort afforded to drivers is poor.

LOS F describes unstable flow. Such conditions exist within queues forming behind bottlenecks. Breakdowns occur for a number of reasons:

- Traffic incidents can temporarily reduce the capacity of a short segment, so that the number of vehicles arriving at a point is greater than the number of vehicles that can move through it.
- Points of recurring congestion, such as merge or weaving segments and lane drops, experience very high demand in which the number of vehicles arriving is greater than the number of vehicles that can be discharged.
- In analyses using forecast volumes, the projected flow rate can exceed the estimated capacity of a given location.

In all cases, breakdown occurs when the ratio of existing demand to actual capacity, or of forecast demand to estimated capacity, exceeds 1.00. LOS F operations within a queue are the result of a breakdown or bottleneck at a downstream point. In practical terms, the point of the breakdown has a d/c ratio greater than 1.00 and is also labeled LOS F, although actual operations at the breakdown point and immediately downstream may actually reflect LOS E conditions. Whenever queues due to a breakdown exist, they have the potential to extend upstream for considerable distances. In that case, the upstream conditions (in the queue) will likely operate at LOS F speeds and densities, even if the segment-level predictions are LOS E or better. Therefore, for accurate estimation of the operational performance of these queue spillback effects, a freeway facility analysis should be conducted by using the procedure in Chapter 10 whenever one or more segment demands exceed capacity.

Oversaturated conditions are represented by LOS F.

LOS Criteria

A basic freeway or multilane highway segment can be characterized by three performance measures: density in passenger cars per mile per lane, space mean speed in miles per hour, and the ratio of demand flow rate to capacity (v/c). Each of these measures is an indication of how well traffic is being accommodated by the basic freeway segment.

Because speed is constant through a broad range of flows and the v/c ratio is not directly discernible to road users (except at capacity), the service measure for basic freeway and multilane highway segments is density. Exhibit 12-15 shows the criteria.

LOS	Density (pc/mi/ln)
A	≤ 11
B	$> 11-18$
C	$> 18-26$
D	$> 26-35$
E	$> 35-45$
F	Demand exceeds capacity OR density > 45

Exhibit 12-15

LOS Criteria for Basic Freeway and Multilane Highway Segments

The LOS thresholds for basic freeway and multilane highway segments are the same for urban and rural locations, as defined by the FHWA smoothed or adjusted urbanized boundaries (8). However, note that a freeway facilities analysis (Chapter 10) defines different LOS thresholds for urban and rural *facilities*.

For all levels of service, the density boundaries on basic freeway segments are the same as those for multilane highways. Traffic characteristics are such that the maximum flow rates at any given LOS are lower on multilane highways than on similar basic freeway segments.

The specification of maximum densities for LOS A to D is based on the collective professional judgment of the members of the Transportation Research Board's Committee on Highway Capacity and Quality of Service. The upper value shown for LOS E (45 pc/mi/ln) is the maximum density at which sustained flows at capacity are expected to occur. In effect, as indicated in the speed-flow curves of Exhibit 12-7, when a density of 45 pc/mi/ln is reached, flow is at capacity, and the v/c ratio is 1.00.

In the application of this chapter's methodology, however, LOS F is identified when demand exceeds capacity because the analytical methodology *does not allow* the determination of density when demand exceeds capacity. Although the density will be greater than 45 pc/h/ln, the methodology of Chapter 10, Freeway Facilities Core Methodology, must be applied to determine a more precise density for such cases.

Exhibit 12-16 illustrates the range of densities for a given LOS on the base speed-flow curves for basic freeway segments. On a speed-flow plot, density is a line of constant slope starting at the origin. The LOS boundaries were defined to produce reasonable ranges for each LOS letter. Exhibit 12-17 shows the same relationships applied to multilane highway segments. The two dashed lines in the latter exhibit correspond to speed-flow relationships that were extrapolated from other results but that have not been calibrated from field data.

Exhibit 12-16

LOS Criteria and Speed-Flow
Curves for Basic Freeway
Segments

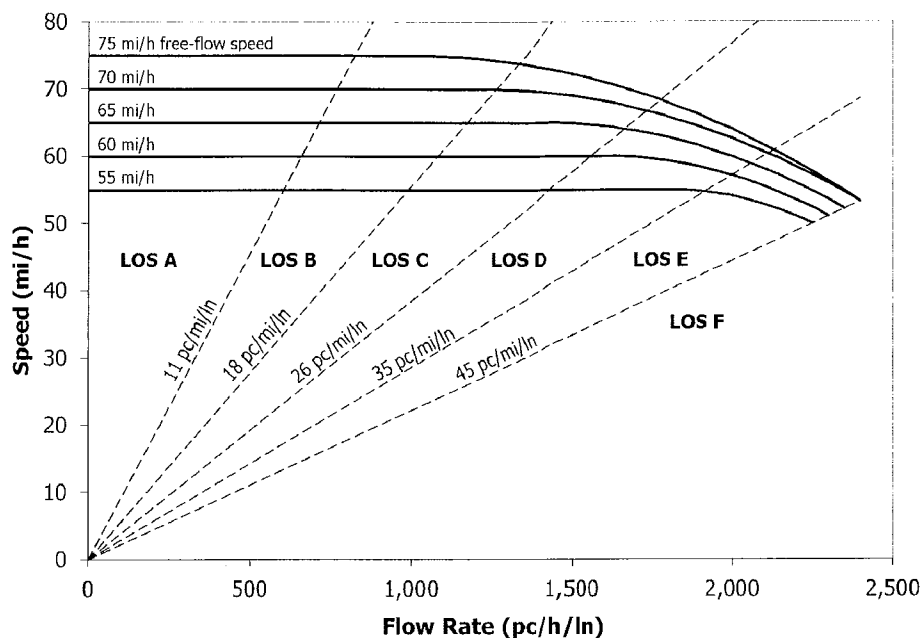
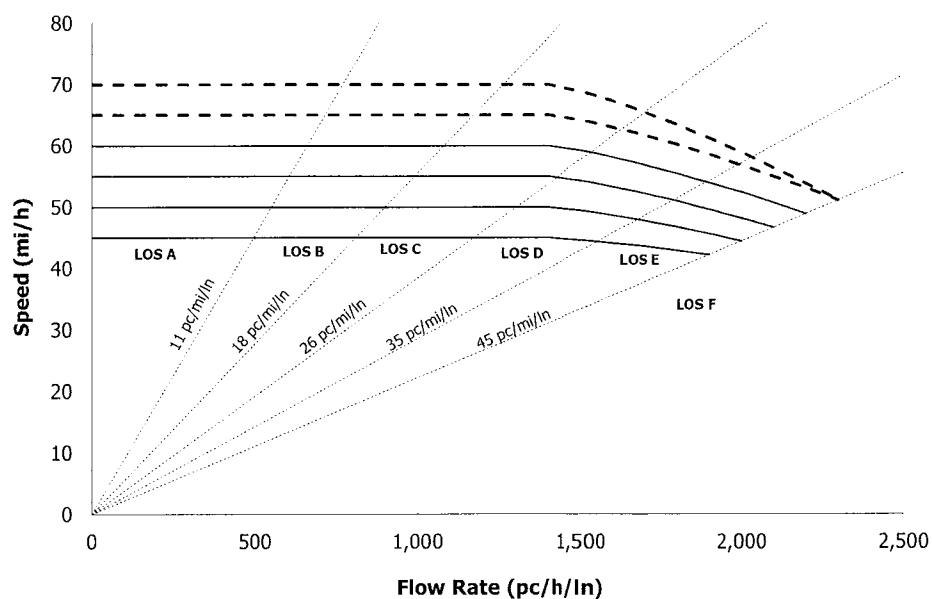


Exhibit 12-17

LOS Criteria and Speed-Flow
Curves for Multilane Highway
Segments



Note: Dashed curves are extrapolated and not based on field data.

3. MOTORIZED VEHICLE CORE METHODOLOGY

This chapter's methodology can be used to analyze the capacity, LOS, and lane requirements of basic freeway or multilane highway segments and the effects of design features on their performance. The methodology is based on the results of an NCHRP study (4), which has been partially updated (5). A number of significant publications were also used in the development of the methodology (6, 7, 9–17).

SCOPE OF THE METHODOLOGY

The methodology described in this section is applicable to general purpose uninterrupted-flow, undersaturated basic freeway and multilane segments. Oversaturated conditions on basic freeway segments can be analyzed with the method described in Chapter 10, Freeway Facilities Core Methodology. Extensions of the methodology described in Section 4 address basic managed lane segments and bicycle LOS on multilane highways. Chapter 26, Freeway and Highway Segments: Supplemental, presents a method to analyze freeway operations on segments with significant truck presence, a prolonged single upgrade, or both.

Spatial and Temporal Limits

Determining capacity or LOS requires uniform traffic and roadway conditions on the analysis segment. Thus, any point where roadway or traffic conditions change must mark a boundary of the analysis segment.

At every ramp–freeway (or ramp–multilane highway) junction, the demand volume changes as some vehicles enter or leave the traffic stream. Thus, any ramp junction should mark a boundary between adjacent basic freeway or multilane highway segments.

In addition to ramp–freeway junctions, the following conditions generally dictate that a boundary be established between basic freeway or multilane highway segments:

- Change in the number of lanes (cross section);
- Changes in lane width or lateral clearance;
- Grade change of 2% or more on a specific or composite grade;
- Change in terrain category (for general terrain segments);
- Presence of a traffic signal, STOP sign, or roundabout along a multilane highway;
- Significant change in the access point density or total ramp density;
- Presence of a bottleneck condition;
- Change in posted speed limit; or
- Presence of an access point at which a significant number or percentage of vehicles enters or leaves a multilane highway.

Ramp junctions, grade changes of 2% or more, changes in the freeway's geometric characteristics, and changes in speed limit are some of the conditions dictating establishment of basic freeway segment or multilane highway boundaries.

The last item in this list is not directly involved in the analysis of a basic freeway or multilane highway segment but would probably reflect changes in ramp or access point density or other features.

The analysis period for any freeway or multilane highway analysis is generally the peak 15-min period within the peak hour. Any 15-min period can be analyzed, however.

If demand volumes are used, demand flow rates are estimated through use of the peak hour factor (PHF). When 15-min volumes are measured directly, the analysis period within the hour that has the highest volumes is selected, and flow rates are the 15-min volumes multiplied by 4. For subsequent computations in the methodology, the PHF is set to 1.00.

Performance Measures

The core motorized vehicle methodology generates the following performance measures:

- Capacity,
- FFS,
- Demand- and volume-to-capacity ratios,
- Space mean speed,
- Average density, and
- Motorized vehicle LOS.

Limitations of the Methodology

This chapter's methodologies for basic freeway segments and multilane highways do not apply to or take into account (without modification by the analyst) the following:

- Lane controls (to restrict lane changing);
- Extended bridge and tunnel segments;
- Segments near a toll plaza;
- Facilities with a FFS more than 75 mi/h for basic freeway segments or more than 70 mi/h for multilane highways;
- Facilities with a base FFS less than 55 mi/h for freeways and less than 45 mi/h for multilane highways, although lower FFS values can be achieved for freeway segments by calibrating a SAF;
- Posted speed limit and enforcement practices;
- Presence of intelligent transportation systems (ITS) related to vehicle or driver guidance;
- Capacity-enhancing effects of ramp metering;
- The influence of downstream queuing on a segment;
- Operational effects of oversaturated conditions; and
- Operational effects of construction operations.

Active traffic and demand management measures for freeways discussed in Chapter 37 consist of the following:

- Ramp metering,
- Congestion pricing,
- Traveler information systems,
- Dynamic lane and shoulder management,
- Speed harmonization,
- Incident management, and
- Work zone traffic management.

Many of these strategies can be evaluated with methodologies in Chapter 11.

The last four items in the list of limitations above are addressed in a freeway facility analysis context, as described in Chapter 10. The following are additional limitations for this chapter's multilane highway methodology:

- The effect of lane drops and lane additions at the beginning or end of multilane highway segments;
- Possible queuing impacts when a multilane highway segment transitions to a two-lane highway segment;
- The negative impacts of poor weather conditions, traffic accidents or incidents, railroad crossings, or construction operations on multilane highways;
- Differences between various types of median barriers and the difference between the impacts of a median barrier and a TWLTL;
- Significant presence of on-highway parking;
- Presence of bus stops that have significant use; and
- Significant pedestrian activity.

The last three factors are more representative of an urban or suburban arterial, but they may also exist on multilane highway facilities with more than 2 mi between traffic signals. When these factors are present on uninterrupted-flow segments of multilane highways, the methodology does not deal with their impact on flow. In addition, this methodology cannot be applied to highways with a total of three lanes in both directions, which should be analyzed as two-lane highways with periodic passing lanes by using the methods of Chapter 15.

Uninterrupted-flow multilane highway facilities that allow access solely through a system of on-ramps and off-ramps from grade separations or service roads should be analyzed as freeways. Note that some ramp access or egress points may be present on a multilane highway where most access or egress points are at-grade junctions of some type.

To address most of the limitations listed above, the analyst would have to utilize alternative tools or draw on other research information and develop special-purpose modifications of this methodology. Operational effects of oversaturated conditions, incidents, work zones, and weather and lighting conditions can be evaluated with the methodology of Chapter 10 and adjustment factors for capacity and FFS found in Chapter 11. Operational effects of active traffic and demand management (ATDM) measures can be evaluated by using the procedures in Chapter 11, Freeway Reliability Analysis. A broader overview of ATDM strategies is presented in Chapter 37, ATDM: Supplemental.

Alternative Tools

Strengths of HCM Procedures

This chapter's procedures were developed on the basis of extensive research supported by a significant quantity of field data. They have evolved over a number of years and represent an expert consensus.

Specific strengths of the HCM basic freeway and multilane highway segment methodology include the following:

Uninterrupted-flow multilane highway facilities that allow access solely through a system of on-ramps and off-ramps from grade separations or service roads should be analyzed as freeways.

The HCM methodology provides FFS as an output, incorporates geometric characteristics, provides explicit capacity estimates, and produces a single deterministic estimate of traffic density.

Deterministic models yield the same results for the same inputs each time they are implemented; stochastic models incorporate statistical variability. The same inputs yield different results in each use. For such models, the average result of X usages is taken as output.

- It provides a detailed methodology for measuring or estimating FFS. This methodology is based on various geometric characteristics. In simulation packages, FFS (or an equivalent, such as desired speed) is usually an input.
- It considers geometric characteristics (such as lane widths), which are rarely, if ever, incorporated into simulation algorithms.
- It provides explicit capacity estimates. Simulation packages do not provide capacity estimates directly. Capacity estimates can only be obtained from simulators through multiple runs with oversaturated conditions. The user can modify simulated capacities by modifying specific input values such as the minimum acceptable headway.
- It produces a single deterministic estimate of traffic density, which is important for some purposes such as development impact review.

Limitations of HCM Procedures That Might Be Addressed by Alternative Tools

Basic freeway segments can be analyzed by using a variety of stochastic and deterministic simulation packages that include freeways. These packages can be useful in analyzing the extent of congestion when there are failures within the simulated facility range and when interaction with other freeway segments and other facilities is present. Less is known about the ability of simulation models to characterize multilane highway operations.

Additional Features and Performance Measures Available from Alternative Tools

This chapter provides a methodology for estimating the capacity, speed, and density of a basic freeway segment, given the segment's traffic demand and characteristics. Alternative tools offer additional performance measures, including delay, stops, queue lengths, fuel consumption, pollution, and operating costs.

REQUIRED DATA AND SOURCES

The analysis of a basic freeway or multilane highway segment requires details concerning the geometric characteristics of the segment and the demand characteristics of the users of the segment. Exhibit 12-18 shows the data that are required to conduct an operational analysis and suggested default values when site-specific data are unavailable (18). The analyst may replace the default values of Exhibit 12-18 with defaults that have been locally calibrated.

The exhibit further distinguishes between urban and rural conditions for certain defaults. The classification of a facility into urban and rural is made on the basis of the FHWA smoothed or adjusted urbanized boundary definition (8), which in turn is derived from Census data.

Required Data and Units	Potential Data Source(s)	Suggested Default Value
<i>Geometric Data—Basic Freeway Segments</i>		
Free-flow speed (mi/h)	Direct speed measurements, estimate from design speed or speed limit	Base free-flow speed: speed limit + 5 mi/h (range 55–75 mi/h)
Number of mainline freeway lanes in one direction (ln)	Road inventory, aerial photo	At least 2
Lane width (ft)	Road inventory, aerial photo	12 ft (range 10–12 ft)
Right-side lateral clearance (ft)	Road inventory, aerial photo	10 ft (range 0–10 ft)
Total ramp density (ramps/mi)	Road inventory, aerial photo	Must be provided (range 0–6 ramps/mi)
Terrain type (level, rolling, specific grade)	Design plans, analyst judgment	Must be provided
<i>Geometric Data—Multilane Highway Segments</i>		
Free-flow speed (mi/h)	Direct speed measurements, estimate from design speed or speed limit	Base free-flow speed: speed limit + 5 mi/h (range 45–70 mi/h)
Number of mainline freeway lanes (one direction)	Road inventory, aerial photo	At least 2
Lane width (ft)	Road inventory, aerial photo	12 ft (range 10–12 ft)
Right-side lateral clearance (ft)	Road inventory, aerial photo	6 ft (range 0–6 ft)
Median (left-side) lateral clearance (ft)	Road inventory, aerial photo	6 ft (range 0–6 ft)
Access point density (points/mi)	Road inventory, aerial photo	8 access points/mi (rural) 16 access points/mi (low-density suburban) 25 access points/mi (high-density suburban)
Terrain type (level, rolling, specific grade)	Design plans, analyst judgment	Must be provided
Median type (divided, undivided, TWLTL)	Road inventory, aerial photo	Must be provided
<i>Demand Data—Basic Freeway and Multilane Highway Segments</i>		
Hourly demand volume (veh/h)	Field data, modeling	Must be provided
Heavy vehicle percentage (%)	Field data	5% (urban) 12% (rural) ^a
Peak hour factor ^b (decimal)	Field data	Basic freeway segments 0.94 Multilane highways 0.95 (urban) or 0.88 (rural)
Driver population, capacity, and free-flow speed adjustment factors	Field data	1.0 (see also Chapter 26)

Notes: **Bold italic** indicates high sensitivity (>20% change) of service measure to the choice of default value. **Bold** indicates moderate sensitivity (10%–20% change) of service measure to the choice of default value. TWLTL = two-way left-turn lane.
^a See Chapter 26 in Volume 4 for state-specific default heavy vehicle percentages and driver population adjustment factors.
^b Moderate to high sensitivity of service measures for very low PHF values. See the discussion in the text. PHF is not required when peak 15-min demand volumes are provided.

Research into the percentage of heavy vehicles on uninterrupted-flow facilities (18) found a wide range of values from state to state. Section 2 of Chapter 26 provides state-specific defaults for heavy vehicle percentage on the basis of data from the 2004 Highway Performance Monitoring System. States or local jurisdictions that have developed their own values may substitute them. Analysts may wish to develop their own default values on the basis of recent data.

Exhibit 12-18

Required Input Data, Potential Data Sources, and Default Values for Basic Freeway and Multilane Highway Segment Automobile Analysis

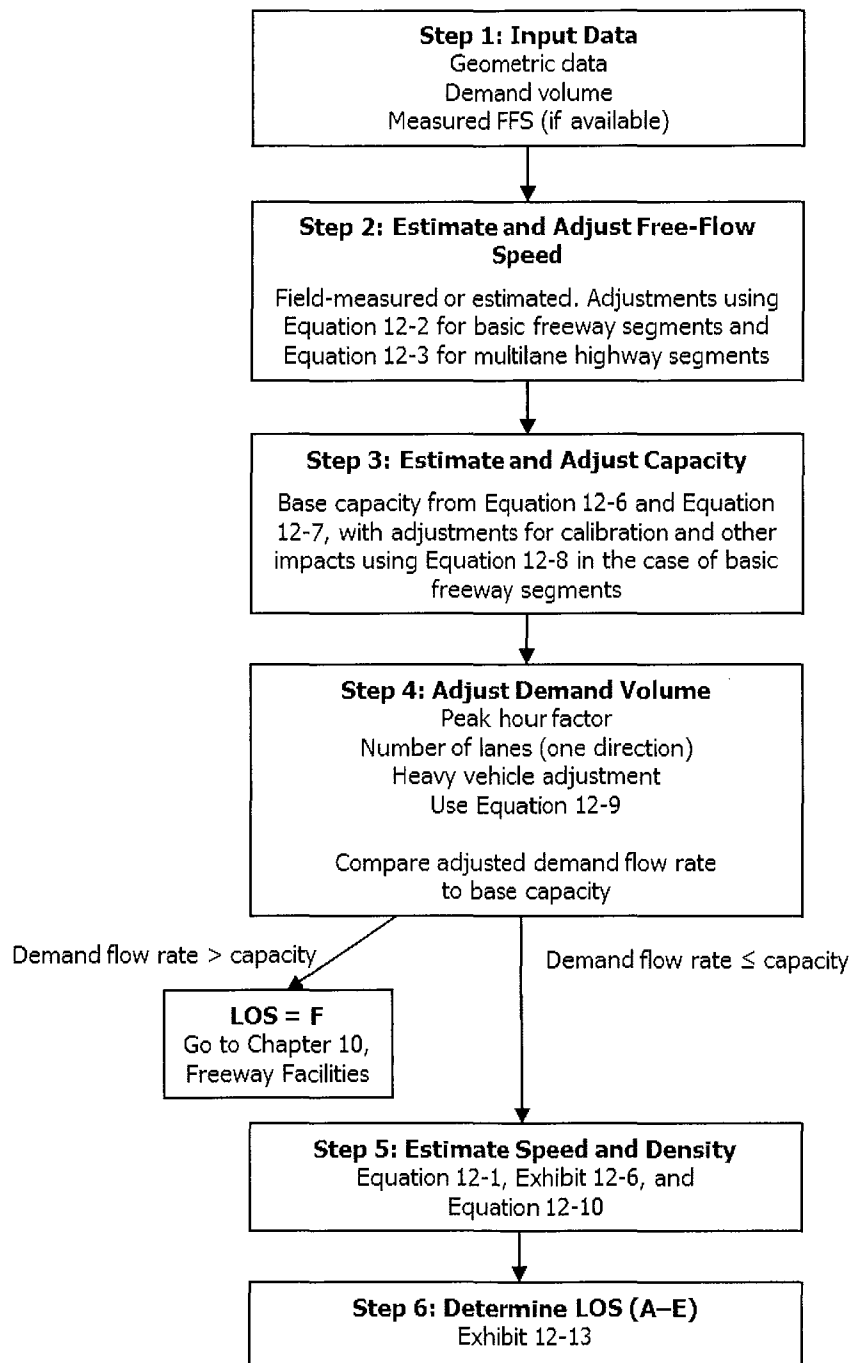
OVERVIEW OF THE METHODOLOGY

Exhibit 12-19 illustrates the basic methodology used in operational analysis. The methodology can also be directly applied to determine the number of lanes required to provide a target LOS for a given demand volume.

Exhibit 12-19

Overview of Operational Analysis Methodology for Basic Freeway and Multilane Highway Segments

Exhibit 12-19 illustrates the methodology for an operational analysis. Other types of analyses are described in Section 5, Applications.



COMPUTATIONAL STEPS

Step 1: Input Data

For a typical operational analysis, as noted previously, the analyst would have to specify (with either site-specific or default values) the demand volume; number and width of lanes; right-side or overall lateral clearance; total ramp or access point density; percent of heavy vehicles; PHF; terrain; and the driver population, speed, and capacity adjustment factors (if necessary).

Step 2: Estimate and Adjust FFS

FFS can be determined directly from field measurements or can be estimated as described below. Statement of FFS in 5-mi/h increments is no longer necessary. This change is important in accounting for the effect of weather or work zones, which may reduce the value of the base FFS.

Field Measurement of FFS

FFS is the mean speed of passenger cars measured during periods of low to moderate flow (up to 500 pc/h/ln). For a specific freeway or multilane highway segment, average speeds are virtually constant in this range of flow rates. Field measurement of FFS, if possible, is preferable. If the FFS is measured directly, no adjustments are applied to the measured value.

Some freeways may have lower posted speed limits for trucks, which may affect the mixed-flow FFS. In these cases, field studies are recommended, since the FFS estimation methodology below is not sensitive to the posted speed limit or the presence of a high percentage of trucks.

The speed study should be conducted at a location that is representative of the segment at a time when flow rates are less than 1,000 pc/h/ln. The speed study should measure the speeds of all passenger cars or use a systematic sample (e.g., every 10th car in each lane). A sample of at least 100 passenger car speeds should be obtained. Any speed measurement technique that has been found acceptable for other types of traffic engineering applications may be used. Further guidance on the conduct of speed studies is provided in standard traffic engineering publications, such as the *Manual of Transportation Engineering Studies* (16).

Estimating FFS

Basic Freeway Segments

Field measurements for future facilities are not possible, and field measurement may not be possible or practical for all existing facilities. In such cases, the segment's FFS may be estimated by using Equation 12-2, which is based on the physical characteristics of the segment under study:

$$FFS = BFFS - f_{LW} - f_{RLC} - 3.22 \times TRD^{0.84}$$

where

FFS = free-flow speed of the basic freeway segment (mi/h);

$BFFS$ = base FFS for the basic freeway segment (mi/h);

f_{LW} = adjustment for lane width, from Exhibit 12-20 (mi/h);

FFS is the mean speed of passenger cars during periods of low to moderate flow.

Equation 12-2

Equation 12-3

f_{RLC} = adjustment for right-side lateral clearance, from Exhibit 12-21 (mi/h);
and
 TRD = total ramp density (ramps/mi).

Multilane Highway Segments

For multilane highway segments, the FFS can be estimated by using Equation 12-3, which is based on the physical characteristics of the segment under study. It is evident that while the base FFS and the lane width adjustment are shared with the estimation method for basic freeway segments in Equation 12-2, the remaining terms are unique to multilane highway segments:

$$FFS = BFFS - f_{LW} - f_{TLC} - f_M - f_A$$

where

FFS = free-flow speed of the multilane highway segment (mi/h);

$BFFS$ = base FFS for the multilane highway segment (mi/h);

f_{LW} = adjustment for lane width, from Exhibit 12-20 (mi/h);

f_{TLC} = adjustment for total lateral clearance, from Exhibit 12-22 (mi/h);

f_M = adjustment for median type, from Exhibit 12-23 (mi/h); and

f_A = adjustment for access point density, from Exhibit 12-24 (mi/h).

Adjustments to FFS

Base FFS

This methodology covers basic freeway segments with a FFS in the range of 55 to 75 mi/h. The predictive algorithm for FFS therefore starts with a value greater than 75 mi/h, specifically a default base FFS of 75.4 mi/h, which resulted in the most accurate predictions in the underlying research.

The methodology covers multilane highway segments with a FFS in the range of 45 to 70 mi/h. The most significant value in Equation 12-3 is $BFFS$. There is not a great deal of information available to help establish a base value. In one sense, it is like the design speed—it represents the potential FFS based only on the highway's horizontal and vertical alignment, not including the impacts of lane widths, lateral clearances, median type, and access points. The design speed may be used for $BFFS$ if it is available.

Although speed limits are not always uniformly set, $BFFS$ for multilane highways may be estimated, if necessary, as the posted or statutory speed limit plus 5 mi/h for speed limits 50 mi/h and higher and as the speed limit plus 7 mi/h for speed limits less than 50 mi/h.

Adjustment for Lane Width

The base condition for lane width is 12 ft or greater. When the average lane width across all lanes is less than 12 ft, the FFS is negatively affected. Adjustments to reflect the effect of narrower average lane width are shown in Exhibit 12-20.

Average Lane Width (ft)	Reduction in FFS, f_{LW} (mi/h)
≥ 12	0.0
$\geq 11-12$	1.9
$\geq 10-11$	6.6

Exhibit 12-20

Adjustment to FFS for Average Lane Width for Basic Freeway and Multilane Highway Segments

Adjustment for Right Lateral Clearance on Freeway Segments

The base condition for right-side lateral clearance is 6 ft or greater. The lateral clearance is measured from the right edge of the travel lane to the nearest lateral obstruction. Care must be taken in identifying a "lateral obstruction." Some obstructions may be continuous, such as retaining walls, concrete barriers, guardrails, or barrier curbs. Others may be periodic, such as light supports or bridge abutments. In some cases, drivers may become accustomed to certain types of obstructions, and their influence on traffic is often negligible.

Exhibit 12-21 shows the adjustment to FFS due to the existence of obstructions closer than 6 ft from the right travel lane edge. Median clearances of 2 ft or more on the left side of the travel lanes generally have little impact on traffic. No adjustments are available to reflect the presence of left-side lateral obstructions closer than 2 ft from the left travel lane edge. Such situations are rare on modern freeways, except in constrained work zones.

Right-Side Lateral Clearance (ft)	Lanes in One Direction			
	2	3	4	≥ 5
≥ 6	0.0	0.0	0.0	0.0
5	0.6	0.4	0.2	0.1
4	1.2	0.8	0.4	0.2
3	1.8	1.2	0.6	0.3
2	2.4	1.6	0.8	0.4
1	3.0	2.0	1.0	0.5
0	3.6	2.4	1.2	0.6

Note: Interpolate for noninteger values of right-side lateral clearance.

Exhibit 12-21

Adjustment to FFS for Right-Side Lateral Clearance, f_{RLC} (mi/h), for Basic Freeway Segments

The impact of a right-side lateral clearance restriction depends on both the distance to the obstruction and the number of lanes in one direction on the basic freeway segment. A lateral clearance restriction causes vehicles in the right lane to move somewhat to the left. This movement, in turn, affects vehicles in the next lane. As the number of lanes increases, the overall effect on freeway operations decreases.

Adjustment for Total Lateral Clearance on Multilane Highway Segments

The adjustment for total lateral clearance (TLC) on multilane highway segments is based on TLC at the roadside (right side) and at the median (left side). Fixed obstructions with lateral clearance effects include light standards, signs, trees, abutments, bridge rails, traffic barriers, and retaining walls. Standard raised curbs are not considered to be obstructions.

Right-side lateral clearance is measured from the right edge of the travel lanes to the nearest periodic or continuous roadside obstruction. If such obstructions are farther than 6 ft from the edge of the pavement, a value of 6 ft is used.

Clearance restrictions on either the right or the left side of the highway reduce the FFS.

Use 6 ft as the left-side clearance for undivided highways and highways with TWLTLs.

Equation 12-4

$$TLC = LC_R + LC_L$$

where

TLC = total lateral clearance (ft) (maximum value 12 ft),

LC_R = right-side lateral clearance (ft) (maximum value 6 ft), and

LC_L = left-side lateral clearance (ft) (maximum value 6 ft).

Exhibit 12-22 shows the reduction in FFS due to lateral obstructions on the multilane highway.

Exhibit 12-22

Adjustment to FFS for Lateral Clearances for Multilane Highways

Four-Lane Highways		Six-Lane Highways	
TLC (ft)	Reduction in FFS, f_{TLC} (mi/h)	TLC (ft)	Reduction in FFS, f_{TLC} (mi/h)
12	0.0	12	0.0
10	0.4	10	0.4
8	0.9	8	0.9
6	1.3	6	1.3
4	1.8	4	1.7
2	3.6	2	2.8
0	5.4	0	3.9

Note: Interpolation to the nearest 0.1 is recommended.

Adjustment for Type of Median on Multilane Highways

The adjustment for type of median is given in Exhibit 12-23. Undivided multilane highways reduce the FFS by 1.6 mi/h.

The FFS is reduced on undivided highways.

Exhibit 12-23

Adjustment to FFS for Median Type for Multilane Highways

Median Type	Reduction in FFS, f_M (mi/h)
Undivided	1.6
TWLT	0.0
Divided	0.0

Adjustment for Total Ramp Density on Basic Freeway Segments

Equation 12-2 includes a term that accounts for the impact of total ramp density on FFS. Total ramp density is defined as the number of ramps (on and off, one direction) located between 3 mi upstream and 3 mi downstream of the midpoint of the basic freeway segment under study, divided by 6 mi. The total ramp density has been found to be a measure of the impact of merging and diverging vehicles on FFS.

Adjustment for Access Point Density on Multilane Highway Segments

Exhibit 12-24 presents the adjustment to FFS for various levels of access point density. Studies indicate that for each access point per mile, the estimated FFS decreases by approximately 0.25 mi/h, regardless of the type of median.

The number of access points per mile is determined by dividing the total number of access points (i.e., driveways and unsignalized intersections) on the right side of the highway in the direction of travel by the length of the segment in miles. An intersection or driveway should only be included in the count if it influences traffic flow. Access points that go unnoticed by drivers or that have little activity should not be used to determine access point density.

Access Point Density (access points/mi)	Reduction in FFS, f_A (mi/h)
0	0.0
10	2.5
20	5.0
30	7.5
≥40	10.0

Note: Interpolation to the nearest 0.1 is recommended.

Although the calibration of this adjustment did not include one-way multilane highway segments, inclusion of intersection approaches and driveways on both sides of the facility might be appropriate in determining the access point density on one-way segments.

Speed Adjustment Factor for Basic Freeway Segments

The estimated FFS for basic freeway segments can be further adjusted to reflect, for example, effects of inclement weather. In this case, an adjusted free-flow speed FFS_{adj} is calculated by multiplying the FFS by a SAF as shown in Equation 12-5:

$$FFS_{adj} = FFS \times SAF$$

where SAF is the speed adjustment factor. The speed adjustment factor can represent a combination of sources, including weather and work zone effects. Default speed adjustment factors and guidance for how to apply them are found in Chapter 11.

The SAF may also be used to calibrate the estimated FFS for local conditions or other effects that contribute to a reduction in FFS. For example, poor pavement conditions or sun glare may cause drivers to reduce their speeds even under low-volume conditions. The adjusted FFS can be used directly in the speed-flow relationship for basic freeway segments in Exhibit 12-6 to define a continuous speed-flow curve that explicitly considers this adjusted FFS. Finally, the effect of unfamiliar drivers on FFS can also be accounted for by using an adjusted FFS. While the driver population SAF defaults to 1.0 in the base procedure, general guidance for selecting an appropriate SAF to account for this factor is given in Section 4 of Chapter 26.

No adjustment of the speed-flow equation using these $SAFs$ is possible for multilane highway segments, since no empirical research exists for applying these effects on multilane highways.

FFS is reduced as the access point density increases.

Exhibit 12-24
Adjustment to FFS for Access Point Density for Multilane Highways

Equation 12-5

Step 3: Estimate and Adjust Capacity

In this step, the base capacity for the basic freeway or multilane highway segment is estimated. The segment capacity is principally a function of the segment FFS, but it can be adjusted to calibrate the segment for local conditions or to reflect impacts of adverse weather conditions, incidents, or other factors. The base capacity values for basic freeway segments and multilane highway segments are listed in Exhibit 12-4 for various values of FFS. Because of the ability to interpolate between different FFS values, the resulting segment capacities should also be interpolated. Alternatively, the base capacity c for a basic freeway segment (in passenger cars per hour per lane) can be estimated directly with Equation 12-6, while the base capacity for a multilane highway segment can be estimated directly with Equation 12-7:

Equation 12-6

$$c \text{ (basic freeway segment)} = 2,200 + 10 \times (FFS_{adj} - 50)$$

Equation 12-7

$$c \text{ (multilane highway segment)} = 1,900 + 20 \times (FFS_{adj} - 45)$$

where all variables have been previously defined.

The capacities resulting from application of these equations can never exceed the base capacities listed in Exhibit 12-4, which are 2,400 pc/h/ln for basic freeway segments and 2,300 pc/h/ln for multilane highway segments. Similarly, the FFS used in these equations should not exceed 75 mi/h for basic freeway segments or 70 mi/h for multilane highway segments.

Adjustment to Capacity for Local Calibration

The base capacities estimated by using Equation 12-6 and Equation 12-7 are based on ideal conditions and are expressed in units of passenger cars per hour per lane. The presence of a significant proportion of heavy vehicles, especially in combination with grades, will result in a net decrease in the observed capacity when converted to units of vehicles per hour per lane. As a result, sensor-based measurements of freeway capacities (in vehicles per hour per lane) may be significantly less than the base values stated above.

Many factors other than heavy vehicle effects can contribute to a reduction in basic freeway segment capacity. Examples of capacity-reducing effects include the following:

- Capacity adjustment for driver population, which is intended to account for the level of unfamiliar drivers in the traffic stream (see Section 4 of Chapter 26 for additional details);
- Turbulence generated from lane drops between two basic segments;
- Turbulence due to merging, diverging, or weaving maneuvers between two basic segments;
- Capacity reductions due to poor sight distance—for example, due to crest vertical curves or horizontal curves;
- Narrow lane widths or low lateral clearances in addition to the effects on FFS presented in Step 2;
- Travel through tunnels or across bridges;
- Poor pavement conditions; and

- Friction effects due to roadside features and attractions that cause drivers to increase following headways.

In these cases, development of a local estimate of capacity and use of that estimate to calibrate a CAF for the segment under study are highly recommended. In the absence of generalized national data on these capacity-reducing effects, a local calibration study or expert judgment is needed to produce a reasonable estimate of segment performance. A methodology for estimating freeway capacities from sensor data is provided in Section 5 of Chapter 26.

Adjustment to Capacity for Basic Freeway Segments

The capacity of a basic freeway segment may be adjusted further to account for the impacts of adverse weather, driver population, occurrence of traffic incidents, or a combination of such influences. The methodology for making these adjustments is the same as that for other types of freeway segments. CAF defaults are found in Chapter 11, along with additional discussion on how to apply them. For convenience, a brief summary is provided here.

The capacity of a basic freeway segment can be adjusted as shown in Equation 12-8:

$$c_{adj} = c \times CAF$$

Equation 12-8

where

c_{adj} = adjusted capacity of segment (pc/h),

c = base capacity of segment (pc/h), and

CAF = capacity adjustment factor (unitless).

The CAF can have several components, including weather, incident, work zone, driver population, and calibration adjustments. The adjustments for weather and incidents are most commonly applied in the context of a reliability analysis as described in Chapter 11, Freeway Reliability Analysis. If desired, capacity can be adjusted further to account for unfamiliar drivers in the traffic stream. While the default CAF for this effect is set to 1.0, guidance is provided in Section 4 of Chapter 26, where estimates for the CAF based on the composition of the driver population are provided.

No adjustment of the speed-flow equation using these CAFs is possible for multilane highway segments, since no empirical research exists for applying these effects to multilane highways.

Step 4: Adjust Demand Volume

Since the speed-flow curves and parameters of Exhibit 12-6 are based on flow rates in equivalent passenger cars per hour on the basic freeway segment, demand volumes expressed as vehicles per hour under prevailing conditions must be converted to this basis by using Equation 12-9:

$$v_p = \frac{V}{PHF \times N \times f_{HV}}$$

Equation 12-9

where

v_p = demand flow rate under equivalent base conditions (pc/h/ln),

V = demand volume under prevailing conditions (veh/h),

PHF = peak hour factor (decimal),

N = number of lanes in analysis direction (ln), and

f_{HV} = adjustment factor for presence of heavy vehicles (decimal).

Peak Hour Factor

The PHF represents the variation in traffic flow within an hour. Observations of traffic flow consistently indicate that the flow rates found in the peak 15 min within an hour are not sustained throughout the entire hour. The application of the PHF in Equation 12-9 accounts for this phenomenon.

On freeways, typical PHFs range from 0.85 to 0.98 (18). On multilane highways, typical PHFs range from 0.75 to 0.95. Lower values within that range are typical of lower-volume conditions. Higher values within that range are typical of urban and suburban peak-hour conditions. Field data should be used if possible to develop PHFs that represent local conditions.

Adjustment for Heavy Vehicles

All heavy vehicles are classified as SUTs or TTs. Recreational vehicles and buses are treated as SUTs. The heavy vehicle adjustment factor f_{HV} is computed from the combination of the two heavy vehicle classes, which are added to get an overall truck percentage P_T .

Equation 12-10

$$f_{HV} = \frac{1}{1 + P_T(E_T - 1)}$$

where

f_{HV} = heavy vehicle adjustment factor (decimal),

P_T = proportion of SUTs and TTs in traffic stream (decimal), and

E_T = passenger car equivalent of one heavy vehicle in the traffic stream (PCEs).

The adjustment factor is found in a two-step process. First, the PCE for each truck is found for the prevailing conditions under study. These equivalency values represent the number of passenger cars that would use the same amount of freeway capacity as one truck under the prevailing conditions. Second, Equation 12-10 is used to convert the PCE values to the adjustment factor.

The effect of heavy vehicles on traffic flow depends on the terrain and grade conditions on the segment as well as traffic composition. PCEs can be selected for one of two conditions:

- Extended freeway and multilane highway segments in general terrain, or
- Specific upgrades or downgrades.

Each of these conditions is more precisely defined and discussed below. However, research has shown that PCEs should be used mostly for addressing capacity and LOS issues. They provide reasonable results for speeds and densities when the grade is slight or the truck percentage is low. For combinations that include steep grades, high truck percentages, or both, the mixed-flow model described in Chapter 25 (for composite grades) and Chapter 26 (for single grades) is recommended for computing mixed-flow speeds and densities and automobile and truck speeds in the mixed traffic stream.

Equivalents for General Terrain Segments

General terrain refers to extended lengths of freeway and multilane highways containing a number of upgrades and downgrades where no one grade is long enough or steep enough to have a significant impact on the operation of the overall segment. General terrain can be either level or rolling. To determine which of these terrain types applies, each upgrade and downgrade should be considered to be a single grade, even if the grade is not uniform. The total length of the upgrade or downgrade is used with the steepest grade it contains. The categorization of a segment as having either level or rolling terrain is as follows:

- *Level terrain:* Any combination of grades and horizontal or vertical alignment that permits heavy vehicles to maintain the same speed as passenger cars. This type of terrain typically contains short grades of no more than 2%.
- *Rolling terrain:* Any combination of grades and horizontal or vertical alignment that causes heavy vehicles to reduce their speed below those of passenger cars but that does not cause heavy vehicles to operate at crawl speeds for any significant length.

No PCE is provided for mountainous terrain, which is any combination of grades and horizontal and vertical alignment that causes heavy vehicles to operate at crawl speed for significant distances or at frequent intervals. In this case, the mixed-flow model presented in Chapters 25 and 26 must be used to estimate speeds and densities. Exhibit 12-25 gives PCEs for the default mix of trucks under level and rolling terrain conditions.

Passenger Car Equivalent	Terrain Type	
	Level	Rolling
E_T	2.0	3.0

Exhibit 12-25
PCEs for General Terrain
Segments

Equivalents for Specific Upgrades

Freeway and multilane highway segments longer than 0.5 mi with grades between 2% and 3% or longer than 0.25 mi with grades of 3% or greater should be considered as separate segments. Research (19) has revealed that the SUT population on freeways has a median weight-to-horsepower ratio of about 100 lb/hp while the TT population has a median weight-to-horsepower ratio of 150 lb/hp. These values can vary from one setting to another.

Exhibit 12-26 gives specific-segment PCE values for a 30%/70% SUT/TT mix, Exhibit 12-27 gives PCE values for a 50%/50% mix, and Exhibit 12-28 gives PCE values for a 70%/30% mix. The 30% SUT condition occurs more frequently on

rural facilities; the 50% condition occurs more frequently on urban facilities. Exhibit 12-28 is recommended for conditions where the majority of the trucks in the traffic stream are SUTs. Note that for the exhibits, segment lengths for grades above 3.5% are limited to 1 mi, because steeper grades are rarely longer than this in practice.

Exhibit 12-26
PCEs for a Mix of 30% SUTs
and 70% TTs

% Grade	Length (mi)	Percentage of Trucks (%)								
		2%	4%	5%	6%	8%	10%	15%	20%	>25%
-2	0.125	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.375	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.625	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.875	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	1.25	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	1.5	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
0	0.125	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.375	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.625	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.875	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	1.25	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	1.5	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
2	0.125	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.375	3.76	2.96	2.78	2.65	2.48	2.38	2.22	2.14	2.09
	0.625	4.47	3.33	3.08	2.91	2.68	2.54	2.34	2.23	2.17
	0.875	4.80	3.50	3.22	3.03	2.77	2.61	2.39	2.28	2.21
	1.25	5.00	3.60	3.30	3.09	2.83	2.66	2.42	2.30	2.23
	1.5	5.04	3.62	3.32	3.11	2.84	2.67	2.43	2.31	2.23
2.5	0.125	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.375	4.11	3.14	2.93	2.78	2.58	2.46	2.28	2.19	2.13
	0.625	5.04	3.62	3.32	3.11	2.84	2.67	2.43	2.31	2.23
	0.875	5.48	3.85	3.51	3.27	2.96	2.77	2.50	2.36	2.28
	1.25	5.73	3.98	3.61	3.36	3.03	2.83	2.54	2.40	2.31
	1.5	5.80	4.02	3.64	3.38	3.05	2.84	2.55	2.41	2.32
3.5	0.125	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.375	4.88	3.54	3.25	3.05	2.80	2.63	2.41	2.29	2.22
	0.625	6.34	4.30	3.87	3.58	3.20	2.97	2.64	2.48	2.38
	0.875	7.03	4.66	4.16	3.83	3.39	3.12	2.76	2.57	2.46
	1.25	7.44	4.87	4.33	3.97	3.50	3.22	2.82	2.62	2.50
	1.5	7.53	4.92	4.38	4.01	3.53	3.24	2.84	2.63	2.51
4.5	0.125	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.375	5.80	4.02	3.64	3.38	3.05	2.84	2.55	2.41	2.32
	0.625	7.90	5.11	4.53	4.14	3.63	3.32	2.90	2.68	2.55
	0.875	8.91	5.64	4.96	4.50	3.92	3.56	3.07	2.82	2.67
	1	9.19	5.78	5.08	4.60	3.99	3.62	3.11	2.85	2.70
	1.5	11.60	7.04	6.11	5.47	4.67	4.18	3.51	3.17	2.97
5.5	0.125	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.375	6.87	4.58	4.10	3.77	3.35	3.09	2.73	2.55	2.44
	0.625	9.78	6.09	5.33	4.82	4.16	3.76	3.21	2.93	2.77
	0.875	11.20	6.83	5.94	5.33	4.56	4.09	3.45	3.12	2.93
	1	11.60	7.04	6.11	5.47	4.67	4.18	3.51	3.17	2.97
	1.5	13.02	7.78	6.71	5.99	5.07	4.51	3.75	3.37	3.14
6	0.125	2.62	2.37	2.30	2.24	2.17	2.12	2.04	1.99	1.97
	0.375	7.48	4.90	4.36	3.99	3.52	3.23	2.83	2.63	2.51
	0.625	10.87	6.66	5.79	5.21	4.46	4.01	3.39	3.08	2.89
	0.875	12.54	7.54	6.51	5.81	4.94	4.40	3.67	3.30	3.08
	1	13.02	7.78	6.71	5.99	5.07	4.51	3.75	3.37	3.14
	1.5	13.02	7.78	6.71	5.99	5.07	4.51	3.75	3.37	3.14

Note: Interpolation in the exhibit is permitted.

% Grade	Length (mi)	Percentage of Trucks (%)								
		2%	4%	5%	6%	8%	10%	15%	20%	>25%
-2	0.125	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.375	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.625	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.875	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	1.25	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	1.5	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
0	0.125	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.375	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.625	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.875	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	1.25	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	1.5	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
2	0.125	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.375	3.76	2.95	2.77	2.64	2.47	2.36	2.20	2.11	2.06
	0.625	4.32	3.24	3.01	2.84	2.63	2.49	2.29	2.19	2.12
	0.875	4.57	3.37	3.11	2.93	2.70	2.55	2.33	2.22	2.15
	1.25	4.71	3.45	3.17	2.99	2.74	2.58	2.36	2.24	2.17
	1.5	4.74	3.47	3.19	3.00	2.75	2.59	2.36	2.24	2.17
2.5	0.125	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.375	4.10	3.13	2.92	2.77	2.57	2.44	2.26	2.16	2.10
	0.625	4.84	3.52	3.23	3.03	2.77	2.61	2.38	2.26	2.18
	0.875	5.17	3.69	3.37	3.15	2.87	2.69	2.43	2.30	2.22
	1.25	5.36	3.79	3.45	3.22	2.92	2.73	2.47	2.33	2.24
	1.5	5.40	3.81	3.47	3.24	2.93	2.74	2.47	2.33	2.25
3.5	0.125	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.375	4.89	3.54	3.25	3.05	2.79	2.62	2.39	2.26	2.19
	0.625	6.05	4.15	3.75	3.47	3.11	2.89	2.58	2.42	2.32
	0.875	6.58	4.43	3.97	3.66	3.26	3.01	2.67	2.49	2.39
	1.25	6.88	4.58	4.10	3.77	3.35	3.09	2.72	2.53	2.42
	1.5	6.95	4.62	4.13	3.80	3.37	3.10	2.73	2.54	2.43
4.5	0.125	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.375	5.83	4.03	3.65	3.39	3.05	2.84	2.55	2.39	2.30
	0.625	7.53	4.92	4.38	4.01	3.53	3.24	2.83	2.62	2.50
	0.875	8.32	5.34	4.72	4.29	3.75	3.42	2.97	2.73	2.59
	1	8.53	5.45	4.81	4.37	3.81	3.47	3.00	2.76	2.62
5.5	0.125	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.375	6.97	4.63	4.14	3.81	3.38	3.11	2.74	2.55	2.43
	0.625	9.37	5.89	5.16	4.68	4.05	3.67	3.14	2.88	2.72
	0.875	10.49	6.48	5.65	5.09	4.37	3.93	3.34	3.03	2.85
	1	10.80	6.64	5.78	5.20	4.46	4.01	3.39	3.08	2.89
6	0.125	2.67	2.38	2.31	2.25	2.16	2.11	2.02	1.97	1.93
	0.375	7.64	4.98	4.43	4.05	3.56	3.26	2.85	2.64	2.51
	0.625	10.45	6.45	5.63	5.07	4.36	3.92	3.33	3.03	2.85
	0.875	11.78	7.16	6.20	5.56	4.74	4.24	3.56	3.22	3.01
	1	12.15	7.35	6.36	5.69	4.85	4.33	3.62	3.27	3.05

Note: Interpolation in the exhibit is permitted.

Exhibit 12-27

PCEs for a Mix of 50% SUTs and 50% TTs

Exhibit 12-28

PCEs for a Mix of 70% SUTs
and 30% TTs

% Grade	Length (mi)	Percentage of Trucks (%)								
		2%	4%	5%	6%	8%	10%	15%	20%	>25%
-2	0.125	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	0.375	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	0.625	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	0.875	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	1.25	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	1.5	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
0	0.125	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	0.375	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	0.625	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	0.875	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	1.25	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
	1.5	2.39	2.18	2.12	2.07	2.01	1.96	1.89	1.85	1.83
2	0.125	2.67	2.32	2.23	2.17	2.08	2.03	1.94	1.89	1.86
	0.375	3.63	2.82	2.64	2.52	2.35	2.25	2.10	2.02	1.97
	0.625	4.12	3.08	2.85	2.69	2.49	2.36	2.18	2.08	2.02
	0.875	4.37	3.21	2.96	2.78	2.56	2.42	2.22	2.11	2.05
	1.25	4.53	3.29	3.02	2.84	2.60	2.45	2.24	2.13	2.07
	1.5	4.58	3.31	3.04	2.86	2.61	2.46	2.25	2.14	2.07
2.5	0.125	2.75	2.36	2.27	2.20	2.11	2.04	1.95	1.90	1.87
	0.375	4.01	3.02	2.80	2.65	2.46	2.33	2.16	2.06	2.01
	0.625	4.66	3.35	3.08	2.88	2.64	2.48	2.26	2.15	2.08
	0.875	4.99	3.52	3.21	3.00	2.73	2.56	2.32	2.19	2.12
	1.25	5.20	3.64	3.30	3.08	2.79	2.60	2.35	2.22	2.14
	1.5	5.26	3.67	3.33	3.10	2.80	2.62	2.36	2.23	2.15
3.5	0.125	2.93	2.45	2.34	2.26	2.16	2.09	1.98	1.92	1.89
	0.375	4.86	3.46	3.16	2.96	2.69	2.53	2.30	2.18	2.10
	0.625	5.88	3.99	3.59	3.32	2.98	2.76	2.46	2.31	2.22
	0.875	6.40	4.26	3.81	3.51	3.12	2.88	2.55	2.38	2.28
	1.25	6.74	4.43	3.96	3.63	3.21	2.96	2.60	2.42	2.32
	1.5	6.83	4.48	3.99	3.66	3.24	2.98	2.62	2.44	2.33
4.5	0.125	3.13	2.56	2.43	2.34	2.21	2.13	2.01	1.95	1.91
	0.375	5.88	3.99	3.59	3.32	2.98	2.76	2.46	2.31	2.22
	0.625	7.35	4.75	4.22	3.85	3.39	3.10	2.71	2.51	2.39
	0.875	8.11	5.15	4.54	4.13	3.60	3.27	2.83	2.61	2.47
	1	8.33	5.27	4.63	4.21	3.66	3.33	2.87	2.64	2.50
5.5	0.125	3.37	2.69	2.53	2.42	2.28	2.19	2.05	1.98	1.94
	0.375	7.09	4.62	4.11	3.76	3.31	3.04	2.66	2.47	2.36
	0.625	9.13	5.68	4.97	4.49	3.88	3.51	3.00	2.74	2.59
	0.875	10.21	6.24	5.43	4.88	4.18	3.76	3.18	2.89	2.71
	1	10.52	6.41	5.57	5.00	4.27	3.83	3.24	2.93	2.75
6	0.125	3.51	2.76	2.59	2.47	2.32	2.22	2.08	2.00	1.95
	0.375	7.78	4.98	4.40	4.01	3.51	3.20	2.78	2.56	2.44
	0.625	10.17	6.23	5.42	4.87	4.17	3.75	3.18	2.88	2.71
	0.875	11.43	6.88	5.95	5.32	4.53	4.04	3.39	3.06	2.86
	1	11.81	7.08	6.11	5.46	4.64	4.13	3.45	3.11	2.90

Note: Interpolation in the exhibit is permitted.

The PCE values shown in this chapter have been estimated from simulation. They are also based on generalized analytical equations for the propulsion and resistance characteristics of SUTs and TTs (19). Different models based on more detailed vehicle dynamics simulators (e.g., 20, 21) can produce different results. The PCEs establish an equivalency between the mixed-traffic capacity and the automobile-only capacity. The speeds associated with these PCE values are space mean speeds, and the densities are defined over the length of the segment. As noted previously, in evaluating composite grades, steep single grades, very high truck percentages, or a combination, the appropriate mixed-flow model from

Chapter 25 (composite grades) or Chapter 26 (single grades) is recommended in lieu of applying PCEs.

Check for LOS F

At this point, the demand volume has been converted to a demand flow rate in passenger cars per hour per lane under equivalent base conditions. This demand rate must be compared with the base capacity of the basic freeway or multilane highway segment (see Exhibit 12-4).

If demand exceeds capacity, the LOS is F and a breakdown has been identified. To analyze the impacts of such a breakdown, the Chapter 10 methodology must be used. No further analysis using the present chapter's methodology is possible. If demand is less than or equal to capacity, the analysis continues to Step 5.

Step 5: Estimate Speed and Density

At this point in the methodology, the following have been determined: (a) the FFS and appropriate FFS curve for use in the analysis and (b) the demand flow rate expressed in passenger cars per hour per lane under equivalent base conditions. With this information, the speed and density of the traffic stream may be estimated.

With the equations specified in Exhibit 12-6, the expected mean speed of the traffic stream can be computed. A graphical solution with Exhibit 12-7 can also be performed.

After the speed is estimated, Equation 12-11 is used to estimate the density of the traffic stream:

$$D = \frac{v_p}{S}$$

Equation 12-11

where

D = density (pc/mi/ln),

v_p = demand flow rate (pc/h/ln), and

S = mean speed of traffic stream under base conditions (mi/h).

As has been noted, Equation 12-11 is only used when v_p/c is less than or equal to 1.00. All cases in which this ratio is greater than 1.00 are LOS F. In these cases, the speed S will be outside the range of Exhibit 12-6 and Exhibit 12-7, and no speed can be estimated.

Where LOS F exists, the analyst should consult Chapter 10, which allows an analysis of the time and spatial impacts of a breakdown, including its effects on upstream and downstream segments.

Step 6: Determine LOS

Exhibit 12-15 is entered with the density obtained from Equation 12-11 to determine the expected prevailing LOS.